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Leveraging leadership and digital transformation for sustainable development: Insights from Cambodia's public sector☆,☆☆

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ABSTRACT

This study examines the influence of transformational leadership on digital transformation and sustainable development outcomes in Cambodian public organizations. Utilizing data from 297 public employees and analyzed through Partial Least Squares Structural Equation Modeling (PLS-SEM), the results reveal that transformational leadership significantly impacts digital transformation and sustainable development outcomes. Additionally, digital transformation not only mediates the connection between transformational leadership and sustainable development outcomes but also amplifies the effects of leadership on sustainability. These findings highlight the crucial role of digital transformation in enhancing the impact of leadership on sustainability, offering valuable insights for public sector leaders and policymakers focused on achieving development objectives. This research makes a unique contribution by exploring the intersection of leadership, digitalization, and sustainability in the public sector of a developing country, an area that has received limited attention.

1. Introduction

Cambodia has experienced remarkable socio-economic growth over the past two decades, averaging an annual GDP growth rate of approximately 7 % before the COVID-19 pandemic. This rapid development positions Cambodia as one of Southeast Asia's fastest-growing economies. In alignment with the United Nations Sustainable Development Goals (SDGs) for 2030, the Cambodian government has initiated the Pentagonal Strategy, Phase 1, which aims to promote growth, employment, equity, efficiency, and sustainability. However, the country's trajectory toward these goals faces substantial challenges, including governance issues, institutional reforms, and inequalities in education and healthcare systems. A comprehensive approach is critical to address these systemic challenges and sustain inclusive economic growth. Despite the attention that transformational leadership and digital transformation have received in academic and policy discussions, there is a significant gap in understanding their cross-cultural applicability, especially in the context of developing countries like Cambodia. It is crucial to address this gap to fully harness the potential of these concepts in diverse socio-economic settings. While transformational leadership has been extensively studied in developed nations [1], its role in fostering innovation, motivating public sector employees, and driving digital transformation initiatives within developing countries remains underexplored. Furthermore, the cross-cultural dimensions of transformational leadership are poorly understood, including how it operates in different governance structures and socio-economic settings. The existing literature on public sector leadership in developing contexts, particularly the digital transformation of public services, has yet to fully account for the barriers and opportunities that arise in countries with limited technological infrastructure and different leadership dynamics. Additionally, the current research lacks a cohesive theoretical framework that connects transformational leadership, digital transformation, and sustainable development outcomes. Leadership styles, such as sustainable leadership, ethical leadership, and responsible leadership, have been widely studied in the context of sustainable performance and development. For example, sustainable leadership, which emphasizes long-term strategic thinking, organizational sustainability, and ethical decision-making, has positively impacted organizational performance by aligning business goals with broader societal needs [2,3]. Ethical leadership, which prioritizes fairness, integrity, and accountability, has also been linked to improved organizational culture, employee satisfaction, and sustainable performance [4,5]. Similarly, responsible

☆ **Ethics statement:** Ethics review or permission was not necessary for the study using human volunteers under local regulations or institutional criteria. The questionnaire informed participants not to provide any identification or information in this work. After being adequately informed, participants submitted consent to participate in the study.☆☆ **Informed Consent:** The researcher adhered to the principles of research ethics, such as the regulation of informed consent.

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leadership, which focuses on considering the impact of leadership decisions on various stakeholders and future generations, has gained significant attention in literature as a key driver of sustainable development in both private and public sectors [6,7]. However, despite the wealth of literature on these leadership styles, a significant research gap exists in understanding how these leadership approaches influence sustainable performance in developing countries, especially in digital transformation. These leadership styles have been studied extensively in developed countries, where technological infrastructure and leadership models are more advanced. However, applying these models in developing countries like Cambodia remains underexplored, where the leadership dynamics and socio-economic conditions differ. For instance, leadership in developing countries must adapt to local cultural and institutional contexts by engaging with informal institutions and power structures to address collective action problems [8]. Inspirational leadership can foster persistence and integrity, driving sustainable practices even in challenging environments [9]. However, corruption, weak institutions, and limited technological infrastructure pose significant barriers, requiring leaders to build trust and innovate [10,8]. Sustainable leadership also involves balancing economic growth with environmental responsibility through eco-efficient technologies and responsible supply chains [11], while community involvement remains a key but challenging strategy for development [12]. Previous studies have demonstrated links between transformational leadership and organizational outcomes [13,14]. However, there is little clarity on how leadership influences digital transformation, particularly in the public sector. This gap is especially pertinent in developing countries where socioeconomic, political, and technological challenges present distinct obstacles to digital adoption and innovation. Although digital transformation has been acknowledged as a critical driver of efficiency, transparency, and citizen engagement in public organizations [15], the relationship between leadership and these outcomes remains disconnected from a broader understanding of how these factors contribute to sustainable development. Moreover, most studies examining the intersection of transformational leadership and digital transformation focus on developed nations, where technological infrastructure and leadership models differ significantly from those in developing economies. As a result, the pathways through which transformational leadership influences digital transformation efforts in the public sectors of developing countries, including Cambodia, remain primarily unexamined [16,17]. This is a critical gap because developing nations face unique challenges, such as limited access to digital technologies, inadequate digital skills among the workforce, and varying socio-economic conditions, which impact the effectiveness of leadership and digital transformation strategies. However, addressing this gap presents a significant opportunity to unlock the potential of transformational leadership in driving digital transformation in developing countries, thereby contributing to sustainable development goals. This study seeks to fill this critical void by investigating how transformational leadership shapes digital transformation efforts within Cambodia's public sector and how these efforts, in turn, influence sustainable development outcomes. Specifically, it explores the mediating role of digital transformation in the relationship between leadership and development outcomes, offering a theoretical framework that connects these variables in the context of a developing economy. This study expands on traditional leadership views by considering the potential influence of different leadership styles on sustainable performance and development. It introduces a broader perspective for understanding how leadership impacts digital transformation and sustainable development outcomes. By examining the mechanisms through which transformational leadership drives digital transformation and promotes sustainable development, this study provides insights crucial for policymakers and public sector leaders seeking to implement effective and sustainable digital strategies. This research contributes to the growing body of literature on leadership in the public sector, focusing on the unique challenges and opportunities in developing countries. The findings offer actionable recommendations for improving public sector performance through leadership development, investment in digital infrastructure, and strategic change management, ultimately supporting Cambodia's pursuit of the SDGs.

1.1. Transformational leadership

Transformational leadership has long been regarded as a robust framework for inspiring change, aligning values, and fostering innovation [18]. Built on four key components—idealized influence, inspirational motivation, intellectual stimulation, and individualized consideration [19]—this leadership style has traditionally been celebrated for its adaptability. However, the assumption that transformational leadership naturally evolves to meet the challenges of digital transformation remains largely unexamined. As organizations navigate rapid technological shifts, it is crucial to question whether transformational leadership is sufficient or if new leadership paradigms are required. One of the most pressing concerns in digital leadership is the ethical complexity introduced by AI-driven decision-making and automation. Idealized influence assumes that leaders act as ethical role models, yet the increasing reliance on algorithms and data-driven processes complicates ethical decision-making. Leaders are now expected to balance innovation with concerns about digital surveillance, algorithmic bias, and data privacy—issues that transformational leadership theory does not fully address [20–22]. While some scholars argue that ethical leadership remains a core function of transformational leadership, others question whether leaders can maintain ethical integrity when opaque technological systems mediate their decisions. Similarly, inspirational motivation—a key tenet of transformational leadership—has become more complex in an era of digital disruption. Traditional leadership models assume that motivation stems from a compelling vision, yet digital transformation often brings uncertainty, job displacement, and resistance to change. While studies highlight the role of transformational leaders in fostering innovation [23,22], few studies examine whether these leaders are adequately prepared to address the anxieties and disruptions accompanying technological change. Inspiring employees is no longer enough—leaders must actively mitigate the risks and uncertainties associated with digital transformation. Intellectual stimulation, another pillar of transformational leadership, is essential for fostering innovation and creative problem-solving. However, the literature tends to focus on how leaders encourage innovation rather than how they help employees cope with digital fatigue and resistance to change [24–27]. The pace of technological change indicates that organizations must constantly adapt, yet existing research often assumes a linear progression of innovation rather than addressing the psychological toll of perpetual disruption. Gaps still exist in the literature on how transformational leaders balance innovation with well-being, ensuring employees remain engaged without being overwhelmed. Individualized consideration—the focus on personalized support and mentorship—also faces new challenges in the digital era. Remote work, AI-driven HR systems, and digital communication tools have transformed how leaders engage with employees. While these technologies offer greater flexibility and accessibility, they also risk depersonalizing leadership interactions, making it harder for leaders to maintain deep, personal connections with their teams [28–31]. Existing studies often assume that digital tools enhance leadership effectiveness, but there is little research on whether they might erode the relational aspects of transformational leadership. Beyond corporate settings, transformational leadership is often credited with fostering digital governance and public sector innovation. While scholars argue that transformational leaders can drive digitalization in government [32,33], many studies fail to critically assess whether structural and bureaucratic constraints allow transformational leadership to thrive [34,35]. Public sector leaders may have the vision and motivation to embrace digital transformation, but rigid administrative systems and resistance to change often undermine their ability to implement meaningful reforms. Despite its strengths, transformational leadership cannot be assumed to be an automatic solution to

digital transformation. Ethical dilemmas, digital fatigue, depersonalization, and structural constraints present factual challenges that remain underexplored in the current literature. Digital Transformation

Digital transformation is often positioned as a necessary and beneficial shift in public sector operations, enhancing efficiency, transparency, and citizen engagement [36]. However, this perspective frequently overlooks the complexities and challenges associated with its implementation. Integrating digital technologies does not inherently lead to improved governance or service delivery but instead introduces new risks and barriers that require careful consideration. While digital transformation has the potential to modernize public administration, its success depends on structural readiness, policy frameworks, leadership adaptability, and social inclusivity rather than on technological advancement alone. One of the prevailing assumptions about digital transformation is that technology adoption naturally improves efficiency. Automation and digital platforms are credited with streamlining processes, reducing administrative burdens, and accelerating service delivery [37]. While e-governance initiatives have simplified bureaucratic procedures in many contexts, their effectiveness is contingent on infrastructure readiness, workforce digital literacy, and the seamless integration of systems [38]. In developing countries, persistent challenges such as unreliable internet access, outdated systems, and limited technological resources undermine the efficiency gains expected from digitalization [39,40]. Furthermore, digital inefficiencies can emerge when public sector employees are not adequately trained to navigate new systems, resulting in bottlenecks rather than streamlined processes. Given these technological and institutional limitations, the assumption that digital transformation inherently enhances efficiency must be reconsidered.

Transparency is another widely cited advantage of digital transformation, as digital governance initiatives promote open data access, reduce corruption, and enhance government accountability [41,42]. While open data platforms and digital reporting mechanisms have facilitated greater public access to information, their effectiveness depends on governance structures and regulatory oversight. Without robust accountability frameworks, digital technologies can obscure rather than enhance transparency. The increasing reliance on AI-driven decision-making introduces additional concerns regarding algorithmic bias, data manipulation, and opaque governance structures, potentially reducing trust in digital public services [43]. Moreover, in some cases, digital governance mechanisms have facilitated surveillance and control rather than genuine transparency, reinforcing the need for a critical examination of how digital tools are deployed and regulated. Citizen engagement is another area where digital transformation is often overestimated. While digital platforms, mobile applications, and online portals provide new channels for interaction, their impact on actual governance outcomes remains uncertain [44,45]. The effectiveness of these tools depends on digital access, trust in digital services, and the extent to which public participation translates into meaningful policy changes. Digital engagement mechanisms often create an illusion of involvement without fundamentally altering decision-making processes. Furthermore, digital transformation can deepen existing social inequalities, as underserved populations, including rural communities and lower-income groups, often lack the necessary infrastructure or digital literacy to engage with online government services. When digitalization does not account for these disparities, it risks reinforcing exclusion rather than fostering inclusive governance.

Leadership and organizational culture also play a crucial role in determining the success of digital transformation. While strong leadership is often considered a driving force behind digital initiatives, it is insufficient [46]. Many public sector institutions remain highly bureaucratic and resistant to change, making it difficult for leaders to implement digital reforms effectively [38]. Even when leadership is supportive, challenges such as rigid administrative structures, employee resistance, and insufficient cross-departmental collaboration can slow down digital transformation efforts. Without addressing these structural and cultural barriers, leadership alone cannot drive meaningful and sustainable digital change. Another factor influencing digital transformation is the development of human capital. Many discussions on digital transformation focus on bridging the skills gap through short-term training programs [47,48]. However, digital transformation requires more than technical training—it necessitates a fundamental shift in how public sector employees work, collaborate, and adapt to ongoing technological changes [49]. Governments often invest in digital literacy programs without aligning them with broader institutional reforms, leading to fragmented implementation and inconsistent outcomes. Ensuring a workforce equipped to handle digital transitions requires long-term investment in professional development, continuous learning, and adaptive management practices. Despite its potential benefits, digital transformation should not be viewed as an automatic solution to governance challenges. Its effectiveness is contingent on strong digital infrastructure, inclusive policies, leadership adaptability, and regulatory frameworks that ensure responsible implementation. Digital transformation efforts that focus solely on technology adoption without addressing institutional readiness and societal disparities may fail to achieve meaningful improvements in governance. Future research should critically assess how digital transformation can be designed to be inclusive, equitable, and responsive to public needs rather than assuming that technology-driven changes will inherently lead to positive outcomes.

1.2. Sustainable development outcomes

Sustainable development is often framed as a universal solution to balancing economic growth, social inclusion, and environmental protection [50]. However, the dominant discourse emphasises policy objectives while overlooking the persistent structural barriers that hinder implementation. While the public sector is crucial in sustainability efforts, government interventions alone are often insufficient due to resource constraints, political resistance, and unintended policy consequences [51,52]. Achieving sustainable development requires a more nuanced approach that critically assesses the effectiveness of existing strategies and explores alternative solutions beyond conventional policy frameworks. One of the most persistent contradictions in sustainability efforts is the conflict between economic growth and environmental protection. Governments promote climate action initiatives, yet many sustainability policies struggle to balance long-term environmental goals with short-term economic priorities. Public sector interventions, such as subsidies for renewable energy, can sometimes create market distortions that disproportionately impact developing economies [53,51]. Instead of assuming that policy interventions will seamlessly integrate sustainability into economic structures, a more critical approach is needed to evaluate how economic systems can be restructured to support environmental and social objectives sustainably.

Leadership styles are often discussed as key drivers of sustainable performance, particularly in state-owned enterprises and nonprofit organizations. While transformational and transactional leadership have been linked to increased motivation and innovation, their direct impact on sustainability remains underexplored [54]. Green leadership fosters environmental knowledge-sharing, but there is limited empirical evidence demonstrating its effectiveness in delivering long-term sustainable outcomes [55]. Similarly, while digital leadership is frequently discussed in sustainability literature, its practical influence is largely theoretical and requires further investigation. Leadership alone is not a sufficient mechanism for achieving sustainability, and future research should focus on how institutional structures, policies, and organizational cultures shape leadership effectiveness in sustainability initiatives. Resource management remains a fundamental challenge in sustainable development, requiring governments to balance conservation efforts with economic demands. While sustainable resource policies aim to protect natural assets such as water, forests, and minerals, these policies often lead to trade-offs that disproportionately affect vulnerable

industries and communities [56]. Water conservation initiatives, for instance, must account for the socioeconomic impact on agriculture and industry, highlighting the complexity of policy decisions in sustainability. Corruption, inefficiency, and lack of accountability in public resource management can further undermine sustainability efforts, making it imperative to assess the actual effectiveness of governance mechanisms rather than relying on theoretical policy frameworks. Public sector service provision is frequently positioned as a pillar of sustainable development, particularly in education, healthcare, and infrastructure investment [57]. However, public sector-led sustainability initiatives are not always the most effective approach, as bureaucratic inefficiencies and governance challenges can impede progress. In some instances, private sector initiatives have demonstrated greater efficiency in advancing sustainability goals, raising the need for a critical reassessment of public-private partnerships in sustainability efforts [58]. The assumption that the public sector is the primary driver of sustainability should be re-evaluated in favour of a more balanced model that leverages private-sector innovation and investment alongside government intervention. Measuring sustainable development remains an ongoing challenge due to the limitations of traditional economic and environmental indicators. GDP per capita and the Gini coefficient are widely used to assess economic progress, but they fail to capture the full spectrum of sustainability, including environmental degradation and long-term social well-being [59]. Similarly, while greenhouse gas emissions and biodiversity loss are critical indicators of environmental sustainability, they often provide an incomplete picture by overlooking interconnected issues such as land degradation and water scarcity [60,61]. Political and economic interests can also influence the selection and interpretation of sustainability indicators, leading to misleading conclusions about progress toward sustainability goals [62]. A multidimensional approach that integrates economic, social, and environmental factors is necessary to achieve a more accurate understanding of sustainability outcomes.

Technological innovation is increasingly important in shaping sustainable development outcomes, but its contribution remains complex and context-dependent. Research suggests that technological advancements can facilitate sustainability by improving knowledge-sharing and efficiency, yet many innovations remain inaccessible to low-income economies due to cost barriers and infrastructure limitations [55]. Frugal innovation, which emphasizes affordable and resource-efficient solutions, has emerged as a promising approach to sustainable development, yet it remains underutilized in large-scale policy initiatives [63]. Rather than assuming that technological progress inherently drives sustainability, it is essential to critically assess the conditions under which innovation contributes to equitable and inclusive sustainability outcomes. Sustainable development is an evolving challenge that requires continuous reassessment of existing frameworks and strategies. While policy initiatives and leadership play important roles, they must be evaluated in the context of economic structures, governance challenges, and institutional effectiveness. A critical examination of sustainability efforts demands a move beyond broad theoretical discussions toward actionable strategies that address structural barriers and implementation gaps.

1.3. Hypotheses developments

Studies show that digital transformational leadership positively influences the success of digital transformation initiatives within organizations [64,65]. This leadership style fosters the skills required for successful digital transformation by encouraging behavioral changes and quicker adaptation to new situations [66]. Although most studies support this positive relationship, some nuances have emerged. For instance, one study found that transformational leadership directly influences employee engagement but has no significant indirect effect through workplace digitalization [67]. Additionally, the impact of transformational leadership on adaptive capability in digital transformation projects was found to be weak in one study, suggesting the need for further exploration of adaptive leadership approaches [66]. Most evidence supports the idea that transformational leadership positively influences digital transformation by fostering innovation, engaging employees, and driving organizational change [68]. This leadership style is particularly effective in developing digital strategies, promoting organizational agility, and enhancing overall performance in digitally transforming organizations [64,65,69]. However, this relationship may be complex and context-dependent, highlighting the need for further research to fully understand the mechanisms by which transformational leadership impacts digital transformation across different organizational settings. Therefore, this study proposes the following hypotheses:

H1: Transformational leadership significantly influences digital transformation

Research indicates that digital technologies can enhance environmental sustainability, social progress, and economic development in alignment with the SDGs [70,71]. Digital transformation has been shown to enhance sustainable innovation performance by aligning digital technologies with business strategies [72]. It positively impacts environmental sustainability and is strengthened by training provisions [73]. Digital inclusion has improved EU countries' environmental, social, and governance (ESG) performance [74]. In Chinese firms, digital transformation dramatically enhances the quality of green innovation, especially when executives possess digital expertise and regions have robust intellectual property protection [75]. However, contradictions and challenges remain. While digital transformation unlocks potential for decent work and local economic growth in rural entrepreneurship, it can also exacerbate inequality and disrupt traditional markets, potentially impeding progress on certain SDGs [76]. Additionally, using digital technologies for sustainability can have both positive and negative effects, such as the risk of increased energy consumption and e-waste generation [77]. These findings suggest that digital transformation positively influences sustainable development outcomes when properly implemented. It can enhance resource efficiency, optimize management practices, and drive innovation toward sustainability [78,79]. To maximize the benefits of digitalization, organizations must balance its advantages with the associated risks and ensure that appropriate human resources and internal controls are in place [79]. As digital sustainability strategies evolve, further research is needed to explore how digital transformation can be leveraged for long-term sustainable development. Accordingly, this study proposes that:

H2: Digital transformation significantly influences sustainable development outcomes

Studies have demonstrated that transformational leadership positively influences organizational sustainability, employee performance, and innovation [80–82]. It positively affects teachers' commitment and sustainable teaching performance [80]. Similarly, a transformational leadership climate (TLC) enhances new product development processes and business sustainability [82]. Green transformational leadership is recognized as an effective strategy for achieving environmental goals and motivating followers to surpass expected environmental performance levels [83]. However, contradictory and exciting findings have been reported previously. In developing countries' textile industries, transformational and transactional leadership styles were negatively correlated with environmental sustainability, with transformational leaders potentially reducing environmental expenditures [81]. In some instances, transformational leadership has also been linked to increased nitrogen emissions, emphasizing the complex relationship between leadership style and sustainability outcomes [81]. In conclusion, while transformational leadership generally positively influences sustainable development outcomes, its effectiveness may vary depending on the context and specific sustainability metrics being considered. The development of tools such as the Sustainable Innovation Matrix (SIM) leadership model [84] and the measurement model for evaluating the impact of transformational leadership on sustainable development goals

within transport companies [85] demonstrate ongoing efforts to understand better and leverage this leadership style for sustainable development. Therefore, this study suggests that:

H3: Transformational leadership significantly influences sustainable development outcomes

Studies reveal that digital transformational leadership and organizational agility positively contribute to successful digital transformation [16,64]. Transformational leadership impacts organizational innovation, which is essential for sustainable development [86]. Studies indicate that digital transformation substantially enhances the quality of green innovation within companies, thereby reinforcing the connection between digital transformation and sustainability [75]. However, some studies have reported conflicting findings. Specifically, one study found that digital transformation does not substantially affect organizational performance or the capacity for innovation in the workplace [87]. On the other hand, another study illustrated a positive correlation between digitalization and the successful attainment of SDGs [70]. This divergence underscores the complexity of the relationship and emphasizes the need for additional research to understand these dynamics better. Therefore, transformational leadership is key in driving digital transformation efforts and enhancing sustainability outcomes. However, the effectiveness of this association may be influenced by several factors, including environmental dynamism, organizational culture, and innovation capabilities [72,88]. Therefore, this study suggests that:

H4: Digital transformation mediates the relationship between leadership and sustainable development outcomes

The framework of this study (Fig. 1) investigated how transformational leadership influences digital transformation efforts within public organizations and how these efforts impact sustainable development outcomes.

1.4. Methodology

This study adopted a quantitative research methodology, utilizing a cross-sectional design. A quantitative approach was selected due to its efficacy in hypothesis testing and establishing statistical correlations between variables [89]. By employing a cross-sectional design, the study captured data at a single point in time, effectively offering a snapshot of how transformational leadership, digital transformation, and sustainable development outcomes interact within Cambodia's public sector. The use of structural equation modeling (SEM) was pivotal in analyzing the proposed relationships, with Partial Least Squares SEM (PLS-SEM) being particularly advantageous for investigating intricate connections among multiple variables, including the intermediary role of digital transformation [90]. PLS-SEM is robust for its capacity to manage smaller to medium-sized samples, non-normally distributed data, and complex model structures involving several indicators and latent constructs [91].

1.5. Population and sample

This study focuses on public-sector employees involved in digital initiatives in Cambodia, ranging from frontline staff to senior managers, to explore how transformational leadership influences digital transformation and its impact on sustainable development outcomes. Stratified random sampling ensured a representative sample by dividing the population into subgroups based on factors such as department or job roles. This method enhances the accuracy and generalizability of the findings by reflecting the diversity of experiences within the public sector [92,93]. Stratified sampling is especially relevant, as Cambodia's public sector organizations vary in digital maturity and leadership practices. Data was gathered between June and August 2024, ensuring sufficient time to recruit participants across different departments and government organizations. Selecting an appropriate sample size is crucial to ensuring the reliability and validity of research findings. Before gathering data, a preliminary power analysis was performed, considering a model encompassing three latent and 18 observed variables. With an effect size of 0.3, a statistical power of 0.95, and a significance threshold of 0.05, the guidelines provided by Soper [94] suggest that a minimum of 200 participants is necessary for obtaining statistically solid results. To enhance the robustness of the study and account for incomplete or missing responses, the sample size increased by 50 %. Data was gathered through a structured survey distributed to public sector employees across various departments via social media and telegram working groups, using validated scales to measure transformational leadership, digital transformation, and sustainable development outcomes, ensuring the reliability and validity of the constructs. These procedures ensured the PLS-SEM analysis's robustness and accuracy, enhancing the findings' validity. Consequently, 297 valid responses exceeded the minimum required sample size of 200. The participants had diverse demographic profiles regarding gender, age, and educational attainment. The sample consisted of 51.9 % males and 48.1 % females, reflecting a relatively balanced gender distribution. The most prominent participants were aged between 36 and 52 years (52.2 %), followed by 42.4 % (18–35 years). A small proportion of participants (5.4 %) were between 53 and 64. Regarding educational background, 47.1 % of the participants had obtained a graduate degree (master's), 44.8 % had completed an undergraduate degree (bachelor's), and 8.1 % held postgraduate or doctoral qualifications.

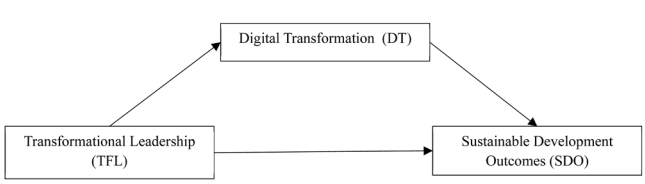

Fig. 1. Conceptual framework.

1.6. Measurement item

The measurement tools used in this research were adapted from well-recognized sources to fit the specific needs of the Cambodian context. A comprehensive scale adaptation process was undertaken to ensure the relevance and cultural appropriateness of the tools. Initially, the study employed a seven-item scale known as the global transformational leadership scale adapted from Carless et al. [95] to evaluate transformational leadership (TFL). In this adaptation, several items were modified to reflect the local leadership styles prevalent in Cambodia, ensuring alignment with cultural norms and expectations. The scale measures various facets of leadership, including vision communication, staff development, support, trust-building, and role modeling. To ensure the psychometric properties of this scale were maintained, a rigorous translation and back-translation process was carried out by bilingual experts fluent in both English and Khmer. This process helped ensure each item's accuracy and cultural relevance, minimizing language biases and ensuring conceptual equivalence with the original scale. Following translation, a pilot test was conducted with a sample of Cambodian leaders, and reliability testing (Cronbach's alpha) was performed to assess internal consistency. The results demonstrated satisfactory reliability and validity in the Cambodian context.

Second, a six-item scale was used to measure the perceived impact of digital transformation on sustainable development outcomes (SDO), evaluating aspects such as economic growth [46,96], social inclusion [97,98], and environmental sustainability [99,100]. This scale was also adapted with input from local experts in digital transformation and sustainable development to reflect Cambodia's unique challenges and opportunities. Similar to the leadership scale, it underwent translation and back-translation processes. The final version was tested for reliability with a sample of Cambodian business and government leaders, yielding satisfactory results. Finally, digital transformation (DT) was adapted from Ly [16] and Nasiri et al. [101] and covered themes of

digitalization efforts, data collection, technology use for expansion, service quality improvement, and information sharing. Items in this study were refined to ensure they captured the digital transformation efforts specific to Cambodia's socio-economic context. In the adaptation process, local industry experts were consulted to refine the relevance of certain items, and a final validation of the scale was conducted using a diverse sample of Cambodian respondents working in various sectors. These adaptations ensured that all scales were culturally and contextually appropriate for the Cambodian setting. The adaptation process was carefully documented, and the validity and reliability of the measures were rigorously assessed to ensure their robustness in this specific cultural and linguistic context.

1.7. Data analysis

Data analysis was structured in several key stages to ensure precision and depth. The process began with a detailed dataset review to address anomalies such as outliers or missing data. Regarding data cleaning procedures, a thorough screening process was conducted before the analysis to ensure the quality of the dataset. Data entry errors were checked by cross-verifying responses with a consistency check algorithm to identify and correct discrepancies. Normality checks were performed on the data using skewness and kurtosis values, with results indicating a normal distribution for most variables. Missing data was addressed by first assessing the pattern of missingness. The amount of missing data was minimal (<5 % of total responses), and missing values were handled using multiple imputations to ensure no information was lost and to reduce any potential bias in the results. The imputation method was chosen because it is appropriate for handling missing data in survey-based studies, mainly when the missingness is random [102]. Outliers were detected using Mahalanobis distance and z-scores, and any extreme cases were carefully reviewed. No significant outliers were identified that would affect the analysis's results. Subsequently, a measurement model for quantitative data analysis was constructed, and its reliability was verified through Cronbach's alpha and composite reliability (CR), ensuring internal consistency.

The assessment then moved to validating the model, focusing first on convergent validity through factor loadings and average variance extracted (AVE), followed by discriminant validity using the Fornell-Larcker criterion [103]. The final stage involved analyzing PLS-SEM with SmartPLS 4, where path coefficients were scrutinized to determine the intensity and direction of variable relationships, directly testing the research hypotheses. This comprehensive method effectively elucidated the research questions, specifically focusing on the Cambodian setting.

1.8. Common method bias

Before proceeding with the factor analysis, the sample's adequacy was evaluated to confirm data validity. The Kaiser-Meyer-Olkin (KMO) test was conducted to this end, requiring a minimum score of 0.6 per the guidelines set by Kaiser and Rice [104]. The obtained KMO score was 0.878, affirming that the data surpassed the threshold. Furthermore, the constructs were scrutinized for multicollinearity before the analysis. This involved calculating the variance inflation factor (VIF) to detect collinearity. Hair et al. [105] suggest a VIF range of 0.20 to 5.0 for PLS-SEM. The VIF values observed ranged from 1.6 to 3.0, indicating the absence of multicollinearity. Harman's single-factor test was also employed to evaluate the potential influence of common method bias (CMB). The test results showed that one factor explained 42.83 % of the variance, below the 50 % cutoff recommended by Podsakoff et al. [106], thus indicating no significant presence of CMB.

1.9. Model assessment

The measurement model's validity and reliability were assessed, with results in Table 1. This table includes statistics on reliability, validity, average variance extracted (AVE), and factor loadings for the variables examined, such as transformational leadership (TFL), digital transformation (DT), and sustainable development outcomes (SDO). Both Cronbach's alpha and composite reliability for these constructs achieved or surpassed the recommended minimum of 0.7, as suggested by Molina et al. [107] (see Table 1). To confirm convergent validity, an examination of the factor loadings for each item in the constructs was undertaken, with all loadings meeting the 0.6 benchmark [108,109], affirming the constructs' validity. Furthermore, the AVE for each construct exceeded 0.5, adhering to the standards set by Hair et al. [103]. The Fornell and Larcker [110] criterion and the heterotrait-monotrait (HTMT) ratio of correlations were utilized to evaluate discriminant validity. This analysis confirmed that the square root of the AVE for each construct was greater than the correlations between constructs (Table 2).

Moreover, the HTMT ratios fell beneath the suggested maximum of 0.90, in line with recommendations by Henseler et al. [111]. As depicted in Table 2, the indicator correlations conformed to the square root of the Average Variance Extracted (AVE), following guidelines from Fornell and Larcker [110]. These findings confirm the good reliability and validity of the tools and the model, thus endorsing the hypotheses presented.

1.10. Structural model

The structural model analysis proceeded once the constructs' validity and reliability were established. We assessed the model's efficacy in forecasting the behavior of endogenous variables using several critical indicators: the coefficient of determination (R^2), cross-validated

Table 1
Factor loadings, reliability, and validity.

Constructs	Loadings
Transformational Leadership (TFL) (Cronbach's Alpha = 0.904, CR = 0.924, AVE = 0.634)	
TFL1-My leader communicates a clear and positive vision of the future	0.784
TFL2-My leader treats staff as individuals, supports and encourages their development	0.844
TFL3-My leader gives encouragement and recognition to staff	0.849
TFL4-My leader fosters trust, involvement, and cooperation among team members	0.784
TFL5-My leader encourages thinking about problems in new ways and questions assumptions	0.758
TFL6-My leader is clear about his/her values and practices what he/she preaches	0.758
TFL7-My leader instills pride and respect in others and inspires me by being highly competent	0.792
Digital transformation (DT) (Cronbach's Alpha = 0.863, CR = 0.901, AVE = 0.645)	
DT1-My organization strives to digitalize everything possible	0.781
DT2-My organization collects vast amounts of data from several sources	0.818
DT3-My organization aspires to use digital technologies for additional activities	0.820
DT4-My organization strives to use digital technology to improve service quality	0.771
DT5-My organization strives towards digital information sharing	0.824
Sustainable Development Outcomes (SDO) (Cronbach's Alpha = 0.871, CR = 0.903, AVE = 0.609)	
SDO1-Digital transformation has improved the efficiency of organizational operations	0.791
SDO2-Digital initiatives have contributed to economic growth in the community	0.794
SDO3-Digital transformation has improved access to services for a broader range of stakeholders	0.812
SDO4-Our organization's digital initiatives have promoted social inclusion	0.797
SDO5-Our organization uses digital technologies to reduce its environmental footprint	0.780
SDO6-Digital transformation has contributed to more sustainable resource management	0.705

Table 2
Discriminant validity-fornell & larcker criterion.

	DT	SDO	TFL
DT	***0.803***		
SDO	0.616	***0.781***	
TFL	0.581	0.624	***0.796***
HTMT			
DT	–		
SDO	0.681		
TFL	0.629	0.681	–

redundancy (Q^2), path coefficients (β), and the significance of the variable relationships. Standardized path coefficients were utilized to gauge the hypotheses' robustness, while the overall model's alignment was verified by examining each structural path's strength, following the approach recommended by Gallardo-Vázquez and Isabel Sánchez-Hernández [112]. The standardized root mean square residual (SRMR), as used in PLS-SEM following Henseler et al. [113], helped minimize model discrepancies. SRMR quantifies the discrepancies between observed and predicted correlations [114], with this study applying SRMR to ascertain the overall model's adequacy. Although PLS-SEM does not strictly define an SRMR threshold [115], a value below 0.10 usually reflects a well-fitting model [116,117], with this study achieving an SRMR of 0.09, indicative of acceptable model alignment. Falk and Miller [118] advocate for an R^2 of at least 0.1 to confirm a path's predictive strength, with R^2 values of 0.75, 0.50, or 0.25 categorized as substantial, moderate, or weak, respectively, for endogenous latent variables [105,113]. The current study registered R^2 values of 0.338 for Digital Transformation and 0.487 for Sustainable Development Outcomes, showcasing adequate predictive capability (refer to Table 3). To evaluate the predictive relevance of endogenous constructs, cross-validated redundancy (Q^2) was employed. A Q^2 value above zero indicates predictive relevance, while a negative value suggests a lack of relevance [119]. The Q^2 values in this analysis were 0.206 for Digital Transformation and 0.279 for Sustainable Development Outcomes, affirming the model's substantial predictive relevance (as shown in Table 3). These findings underscore the interconnected roles of leadership, digital transformation, and sustainability, highlighting the vital importance of transformational leadership in creating a conducive environment for digital innovation and sustainable development practices.

1.11. Direct hypothesis analysis

The results of the hypotheses testing show significant support for all proposed relationships between Transformational Leadership (TFL), Digital Transformation (DT), and Sustainable Development Outcomes (SDO). First, the analysis confirms that Transformational Leadership has a strong positive effect on Digital Transformation ($\beta = 0.581$, $t = 10.805$, $p < 0.001$), indicating that organizations with transformational leaders are more likely to adopt digital transformation initiatives. Furthermore, the findings reveal that Digital Transformation significantly contributes to Sustainable Development Outcomes ($\beta = 0.383$, $t = 8.703$, $p < 0.001$), indicating that advancements in digital transformation practices drive organizational sustainability efforts. This relationship highlights digital innovations and processes' crucial role in achieving sustainable outcomes. Finally, although positive, the direct effect of Transformational Leadership on Sustainable Development Outcomes was significant ($\beta = 0.402$, $t = 7.294$, $p < 0.001$) (Table 3). This finding suggests that transformational leaders directly influence sustainability outcomes.

Moreover, the effect sizes (f^2) were calculated to assess the practical significance of the relationships between constructs in the PLS-SEM analysis (see Table 3). Using Cohen [120] guidelines for interpretation—where an f^2 of 0.02 is considered small, 0.15 is medium, and 0.35 is large—the results indicated that the paths from TFL to DT ($f^2 = 0.511$) and TFL to SDO ($f^2 = 0.208$) demonstrated large and medium effects, respectively. The path from DT to SDO showed a medium effect size ($f^2 = 0.189$). These findings suggest that the relationships between the constructs are not only statistically significant but also practically meaningful, emphasizing the substantial impact of these predictors on their respective dependent constructs.

Table 3
Hypotheses testing.

	Path coefficient	*t*-value	f^2	*p*-value	Decision
H1: TFL -> DT	0.581	10.805	0.511	0.000	Supported
H2: DT -> SDO	0.383	8.703	0.189	0.000	Supported
H3: TFL -> SDO	0.402	7.294	0.208	0.000	Supported
		R^2		Q^2	
DT		0.338		0.206	
SDO		0.487		0.279	

1.12. Mediation analysis

According to Zhao et al. [121], mediation analysis begins by assessing the indirect effect of a * b to determine whether mediation is present. Next, we examine the size of the mediation effect. This study used bootstrapping with 5000 subsamples and bias-corrected 95 % confidence intervals (CI) for more accurate estimates [103,121]. The results demonstrated that the total effect of Transformational Leadership (TFL) on Sustainable Development Outcomes (SDO) was significant and positive ($\beta = 0.624$, $t = 11.962$, $p < 0.001$). When the mediator, Digital Transformation (DT), was introduced, the direct effect of TFL on SDO decreased but remained significant ($\beta = 0.402$, $t = 7.294$, $p < 0.001$). This finding indicates that TFL continues to directly influence SDO even when the mediating effect of DT is considered. The indirect effect of TFL on SDO through DT is statistically significant ($\beta = 0.223$, $t = 6.415$, $p < 0.001$). The confidence interval for this indirect effect ranged from 0.157 to 0.295, and because the interval did not include zero, it confirmed the significance of the mediation effect [103]. Also, the f^2 for the indirect effect of TFL → DT → SDO is approximately 0.189, indicating a medium effect size according to Cohen's guidelines. These findings support H4, which states that DT mediates the relationship between TFL and SDO (see Table 4).

Additionally, the study employed variance accounted for (VAF) to measure the strength of the mediating effect. VAF represents the proportion of the total effect that is mediated. According to Hair et al. [103], the VAF for this model is 36 %, indicating that Digital Transformation (DT) is a complementary partial mediator in the relationship between TFL and SDO. This suggests that while DT explains a significant portion of the relationship, TFL also directly influences SDO.

2. Discussion

This section discusses how these findings align with and contribute to the literature by focusing on comparisons with specific studies from other researchers. This data strongly support the first hypothesis that

Table 4
Mediation analysis.

	Total effects			Direct effects		
	Coefficient	*t*-value	*p*-value	Coefficient	*t*-value	*p*-value
TFL -> SDO	0.624	11.962	0.000	0.402	7.294	0.000
Indirect effects						
Hypotheses	Coefficient	*t*-value	*p*-value	f^2	CI[2.5 %–97.5 %]	VAV
H4: TFL -> DT -> SDO	0.223	6.415	0.000	0.189	.157 - 0.295	36 %

transformational leadership significantly influences digital transformation. This result aligns with those of previous studies, such as that of AlNuaimi et al. [64], who observed a similar positive relationship in government organizations in the UAE. Transformational leaders create environments conducive to digital transformation by fostering innovation and promoting adaptive behaviors. This is further supported by Huang et al. [66], who emphasized that transformational leadership accelerates adaptation to new technologies and processes. However, the complexities of the leadership's role in digital transformation are worth noting. For example, Hooi and Chan [67] found that while transformational leadership positively influences employee engagement, it does not always translate into workplace digitalization. This nuance contrasts with the findings of the current study on the Cambodian public sector, where the effect of leadership on digital transformation is direct and substantial. This difference may be attributed to the organizational context, as public institutions may have distinct motivations and structures compared with private or corporate environments, warranting further research.

The second hypothesis, that digital transformation significantly influences sustainable development outcomes, is also confirmed. This finding is consistent with that of Lin and Mao [72], who show that digital transformation drives sustainability by optimizing business processes and resource efficiency. Similar results were observed in European contexts, where digitalization enhanced environmental, social, and governance (ESG) performance [70]. Although this relationship is generally positive, challenges and contradictions exist in literature. For instance, Prasetyo and Setyadharma [76] highlighted that digital transformation, particularly in rural entrepreneurship, could increase inequality and disrupt traditional markets. These issues underscore the need for a balanced approach to digitalization, ensuring that negative social impacts do not offset sustainability gains. In the Cambodian public sector, where the focus is on achieving broader development goals, the benefits of digital transformation outweigh these risks. However, considering the local context, careful implementation is crucial to maximize these benefits while mitigating potential drawbacks, such as increased energy consumption and e-waste [77].

The results support the third hypothesis that transformational leadership significantly influences sustainable development outcomes and aligns with the broader literature. Mehmood and Rafiq [82] have shown that transformational leadership positively affects sustainability by fostering innovation and encouraging long-term strategic thinking. The findings of this study align with those of Alzoraiki et al. [80], who found that transformational leadership enhances organizational commitment and sustainability, particularly in the education sector. However, some studies have reported a complex relationship. For example, Arun [81] discovered that transformational leadership is associated with decreased environmental sustainability due to reduced environmental expenditure in some developing country industries such as textiles. While this contradiction does not directly apply to the current context, it highlights the importance of understanding industry-specific dynamics. The positive impact of transformational leadership on sustainability in the Cambodian public sector is clear. However, future research should explore whether similar effects are observed in other sectors and contexts.

Finally, this study confirms that digital transformation mediates the relationship between transformational leadership and sustainable development outcomes, thus supporting the fourth hypothesis. The partial mediation found in this study is consistent with the findings of Huang and Lau [75], who noted that digital transformation mediates the relationship between leadership and organizational innovation, thereby enhancing sustainability. The variance accounted for (VAF) of 36 % indicates that, while digital transformation plays a crucial role in transmitting the effects of leadership on sustainability, other factors contribute to this relationship. This suggests that, while transformational leadership directly contributes to sustainable outcomes, its effect is amplified through digital transformation. The Cambodian public sector benefits from this mediation, as leadership-driven digital strategies enhance sustainability efforts, making public organizations more agile and responsive to sustainability challenges.

3. Conclusion

This study contributes significantly to understanding the relationship between transformational leadership, digital transformation, and sustainable development outcomes, particularly within Cambodia's public sector. Empirically testing digital transformation's mediating role sheds light on how public organizations can leverage leadership and technology to drive sustainability. This research fills an important gap in literature, particularly in the context of developing countries, by demonstrating that transformational leadership directly influences both digital transformation and sustainability outcomes. Furthermore, the study highlights that digital transformation partially mediates this relationship, providing new insights into how leadership can effectively impact sustainability. These findings contribute to the theoretical discussion on leadership and sustainability by quantitatively validating that digital transformation is a key mechanism through which leadership influences sustainability.

3.1. Theoretical implications

This study advances the theoretical understanding of how leadership styles, particularly transformational leadership, interact with digital transformation to promote sustainable development. It expands the literature on leadership and sustainability by validating that digital transformation is not merely a technological shift but an essential driver of sustainability outcomes. This work also enriches the understanding of leadership's role in shaping organizational practices that align with long-term sustainability goals. The study's focus on the public sector in developing countries contributes a novel perspective, filling a gap in existing research.

3.2. Practical implications

From a practical standpoint, public sector leaders should prioritize transformational leadership and digital transformation to achieve sustainability objectives effectively. The study emphasizes the need for fostering an organizational culture that encourages innovation, flexibility, and technological adaptation—key components in meeting the UN's Sustainable Development Goals (SDGs). Leaders should integrate digital strategies with broader sustainability efforts to enhance resource efficiency and innovation. Public sector leaders should establish cross-functional task forces dedicated to digital transformation and sustainability to make this actionable. These teams should identify opportunities where technology can drive environmental and social improvements, such as implementing AI-powered resource management or blockchain for transparency in public spending. Public sector managers should also invest in digital infrastructure and focus on developing teams that can adapt to technological changes and align them with sustainability goals. This includes creating targeted training programs that equip employees with the digital skills necessary for implementing sustainable solutions and establishing incentive structures that reward innovative, sustainability-focused digital initiatives. This strategic approach will enable public organizations to meet their environmental and social responsibilities more effectively.

3.3. Policy implications

From a policy perspective, the findings underscore the importance of supporting leadership development and digital transformation within the public sector. To make this a reality, policymakers should design leadership development programs incorporating real-world digital case studies and hands-on training in sustainability-focused technologies,

ensuring that public leaders are prepared to implement change effectively. Policymakers should consider investing in leadership programs focusing on digital and sustainability strategies to prepare leaders for future challenges. Furthermore, policies promoting the adoption of digital technologies in public administration can accelerate the achievement of sustainability goals, especially in developing countries like Cambodia. For a more significant impact, governments should introduce policy incentives—such as tax breaks or grants—for public institutions that successfully integrate digital tools into their sustainability initiatives. Additionally, they should mandate progress tracking through key performance indicators (KPIs) to ensure that digital transformation efforts are aligned with measurable sustainability outcomes.

3.4. Limitations and future research directions

Despite the significant contribution of this study, several limitations must be acknowledged. First, the study was conducted within the context of public organizations in Cambodia, which may limit the generalizability of the findings to other sectors or regions. While the public sector dynamics are central to this study, the results may not fully apply to private sector organizations or those operating in different socio-political environments. Further research could explore the applicability of these findings in countries with varying levels of digital maturity or distinct organizational structures. Second, the cross-sectional nature of the data collection poses a limitation. As the data were collected at a single point in time, it may not capture the long-term effects of transformational leadership and digital transformation on sustainability outcomes. Longitudinal studies would provide a more comprehensive understanding of how these relationships evolve, particularly as digital transformation initiatives mature. Third, the reliance on self-reported data introduces potential biases, such as social desirability bias, where respondents may overstate the effectiveness of leadership or digital transformation initiatives. Future studies could incorporate more objective measures or triangulate data from multiple sources to mitigate such biases. Finally, this study primarily focuses on transformational leadership and its relationship with digital transformation and sustainability. While transformational leadership fosters innovation, other leadership styles, such as transactional or servant leadership, may also play significant roles. Future research could explore the impact of these alternative leadership styles to provide a more comprehensive understanding of how leadership influences organizational processes.

In conclusion, while this study offers valuable insights into the role of transformational leadership and digital transformation in driving sustainable development outcomes in Cambodia's public sector, several limitations should be addressed in future research. Expanding the scope of the study to include other sectors, regions, and leadership styles would enhance the generalizability and depth of understanding of these relationships. Longitudinal studies, more objective measures, and cross-sector comparisons could further clarify the findings' long-term effects and broader applicability. Nonetheless, the study's contributions offer a strong foundation for future research on leadership and sustainability in developing country contexts.

CRediT authorship contribution statement

Bora ly: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] M. Crede, J. Jong, P. Harms, The generalizability of transformational leadership across cultures: a meta-analysis, J. Manag. Psychol. 34 (3) (2019) 139–155, https://doi.org/10.1108/JMP-11-2018-0506.

[2] Y. Liao, Sustainable leadership: a literature review and prospects for future research [Review], Front. Psychol. 13 (2022), https://doi.org/10.3389/fpsyg.2022.1045570.

[3] R. Liu, Z. Yue, A. Ijaz, A. Lutfi, J. Mao, Sustainable business performance: examining the role of green HRM practices, green innovation and responsible leadership through the lens of pro-environmental behavior, Sustainability. 15 (9) (2023) 7317. https://www.mdpi.com/2071-1050/15/9/7317.

[4] V. Marierose, G. Lorena, A. Elaine Joy, B. Mario, Ethical practices and Organizational sustainability: a three dimensional approach, Int. J. Acad. Ind. Res. 3 (4) (2022) 175–194, https://doi.org/10.53378/352954.

[5] M. Nasir, M. Arief, F. Alamsjah, Elidjen, Leveraging environmental management systems and green transformational leadership to enhance employee support for sustainability through ethical behavior, in: 2024 3rd International Conference on Creative Communication and Innovative Technology (ICCIT), 2024.

[6] A. Akbar, S. Chen, M.W. Akhtar, M. Usman, A. Tariq, L. Frija, S.A. Shabbir, Responsible leadership and sustainable outcomes: a multi-study perspective of individual ambidexterity and knowledge sharing culture, Current Issues in Tourism (2024) 1–20, https://doi.org/10.1080/13683500.2024.2394566.

[7] R. Srivastava, Sustainable management strategies in multinational corporations: balancing profitability and social responsibility, J. Adv. Manag. Stud. 1 (3) (2024) 32–35, https://doi.org/10.36676/jams.v1.i3.17.

[8] A.S. Gutterman, Leadership in developing countries, SSRN. 4560325 (2023), https://doi.org/10.2139/ssrn.4560325.

[9] T. Hendayana, A. Purwanto, H. Abbas, The influence of leadership style on ecological behavior mediated by persistence and integrity on the State civil apparatus of the provincial government of the special region of Jakarta, Int. J. Bus., Law, Educ. 5 (2) (2024) 2337–2348, https://doi.org/10.56442/ijble.v5i2.872.

[10] O.A. Ajuwon, A.O. Dagunduro, Strategic leadership and management within non-profit organizations: an in-depth review of practices contributing to sustainable impact, Int. J. Front. Res. Sci. Technol. 3 (1) (2024) 001–017, https://doi.org/10.56355/ijfrst.2024.4.1.0049.

[11] M.K. Meena, A study on sustainable leadership practices: balancing profitability and environmental responsibility, Int. J. Adv. Res. Commerce, Manag. Soc. Sci. (IJARCMSS) 7 (3) (2024) 64–69, https://doi.org/10.62823/IJARCMSS/7.3(II).6871.

[12] D.H. Syahchari, E. Van Zanten, The role of leadership and community involvement in sustainable development, Binus Bus. Rev. 15 (2) (2024) 157–168, https://doi.org/10.21512/bbr.v15i2.10354.

[13] F. Tagscherer, C.-C. Carbon, Leadership for successful digitalization: a literature review on companies' internal and external aspects of digitalization, Sustain. Technol. Entrepreneur. 2 (2) (2023) 100039, https://doi.org/10.1016/j.stae.2023.100039.

[14] B.E. Wright, S.K. Pandey, Transformational leadership in the public sector: does structure matter? J. Public Admin. Res. Theory 20 (1) (2009) 75–89, https://doi.org/10.1093/jopart/mup003.

[15] B. Fuller, A. Bajaba, S. Bajaba, Enhancing and extending the meta-analytic comparison of newer genre leadership forms, Front. Psychol. 13 (2022) 872568, https://doi.org/10.3389/fpsyg.2022.872568.

[16] B. Ly, The interplay of digital transformational leadership, organizational agility, and digital transformation, J. Knowl. Econ. 15 (1) (2024) 4408–4427, https://doi.org/10.1007/s13132-023-01377-8.

[17] E.M.d.M. Souza, P.A. Pietrafesa, Leadership and digital transformation, J. Bus. Admin. Res. (2023), https://doi.org/10.5430/jbar.v12n2p9.

[18] B.M. Bass, R.E. Riggio, Transformational Leadership, Psychology press, 2006.

[19] P.G. Northouse, Leadership: Theory and Practice, Sage publications, 2021.

[20] T.N. Hai, Q.N. Van, M.N. Thi Tuyet, Digital transformation: opportunities and challenges for leaders in the emerging countries in response to COVID-19 pandemic, Emerg. Sci. J. 5 (1) (2021) 21–36, https://doi.org/10.28991/esj-2021-SPER-03.

[21] B. Ly, Transforming commitment into performance: a study of digital transformation in the Cambodian public sector amidst a pandemic, Cogent Bus. Manag. 11 (1) (2024) 2333609, https://doi.org/10.1080/23311975.2024.2333609.

[22] B. Ly, R. Ly, Technostress in times of change: unveiling the impact of leadership styles in Cambodia's public organizations in the wake of COVID-19, Cogent Bus. Manag. 11 (1) (2024) 2331645, https://doi.org/10.1080/23311975.2024.2331645.

[23] C. Ansell, E. Sørensen, J. Torfing, The COVID-19 pandemic as a game changer for public administration and leadership? The need for robust governance responses to turbulent problems, Public Manag. Rev. 23 (7) (2021) 949–960, https://doi.org/10.1080/14719037.2020.1820272.

[24] R. Hisrich, S. Mariusz, Unleashing the creativity of entrepreneurs with digital technologies, Digital Entrepreneur. (2020), https://doi.org/10.1007/978-3-030-53914-6_2.

[25] T. Lê Công, Motivating follower creativity by offering intellectual stimulation, Int. J. Org. Anal. 28 (2019) 817–829, https://doi.org/10.1108/ijoa-06-2019-1799.
[26] S.-S. Moisés, M. Pifarré, C. Andreea, L. Cutillas, F. Enric, The role of digital technologies to promote collaborative creativity in language education, Front. Psychol. 13 (2022), https://doi.org/10.3389/fpsyg.2022.828981.
[27] W. Xiaojing, N. Oi-Lam, C. Zhihao, K.Y.L. Suzannie, Creativity development with problem-based digital making and block-based programming for science, technology, engineering, arts, and mathematics learning in middle school contexts, J. Educ. Comput. Res. 61 (2022) 304–328, https://doi.org/10.1177/07356331221115661.
[28] S. Geng, T. Lijing, N. Ben, F. Yuanyue, C. Li, Knowledge recommendation for workplace learning: a system design and evaluation perspective, Internet Res. 30 (2019) 243–261, https://doi.org/10.1108/intr-07-2018-0336.
[29] T. Leonard, F. Schmiedek, B. Garvin, Developing personalized education: a dynamic framework, Educ. Psychol. Rev. 33 (2020) 863–882, https://doi.org/10.1007/s10648-020-09570-w.
[30] L. Major, G. Francis, T. Maria, The effectiveness of technology-supported personalised learning in low- and middle-income countries: a meta-analysis, Br. J. Educ. Technol. 52 (2021) 1935–1964, https://doi.org/10.1111/BJET.13116.
[31] J. Tomislav, I. Botički, Mobile learning system for enabling collaborative and adaptive pedagogies with modular digital learning contents, J. Comput. Educ. (2019) 1–28, https://doi.org/10.1007/S40692-019-00139-3.
[32] L. Alos-Simo, A.J. Verdu-Jover, J.-M. Gomez-Gras, How transformational leadership facilitates e-business adoption, Ind. Manag. Data Syst. 117 (2) (2017) 382–397, https://doi.org/10.1108/IMDS-01-2016-0038.
[33] H. Dema, H. Hamid, B. Barisan, Transformational Leadership in supporting innovative digital governance, J. Studi Pemerintahan 14 (1) (2023) 69–80, https://doi.org/10.18196/jgp.v14i1.13528.
[34] T.J.J. Brunner, T. Schuster, C. Lehmann, Leadership's long arm: the positive influence of digital leadership on managing technology-driven change over a strengthened service innovation capacity [Original Research], Front. Psychol. 14 (2023), https://doi.org/10.3389/fpsyg.2023.988808.
[35] A. Türk, Digital leadership role in developing business strategy suitable for digital transformation [Original Research], Front. Psychol. 13 (2023), https://doi.org/10.3389/fpsyg.2022.1066180.
[36] G. Vial, Understanding digital transformation: a review and a research agenda, Manag. Digital Transform. (2021) 13–66.
[37] Haffke, I., Kalgovas, B.J., & Benlian, A. (2016). The role of the CIO and the CDO in an organization's Digital transformation.
[38] G.M. Jonathan, K.S. Hailemariam, B.K. Gebremeskel, S.D. Yalew, Public sector digital transformation: challenges for information technology leaders, in: 2021 IEEE 12th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON), 2021.
[39] A.G. Kanaan, A. AL-Hawamleh, A.W. Abulfaraj, H.M. Al-kaseasbeh, A.S. Alorfi, The effect of quality, security and privacy factors on trust and intention to use e-government services, Int. J. Data Network Sci. (2023), https://doi.org/10.5267/j.ijdns.2022.11.004.
[40] H. Zahid, S. Ali, E. Abu-Shanab, H. Muhammad Usama Javed, Determinants of intention to use e-government services: an integrated marketing relation view, Telemat. Inform. 68 (2022) 101778, https://doi.org/10.1016/j.tele.2022.101778.
[41] F. Danielsen, Benefits and challenges of digitalization: an expert study on Norwegian public organizations, in: Proceedings of the 22nd Annual International Conference on Digital Government Research, Omaha, NE, USA, 2021, https://doi.org/10.1145/3463677.3463703.
[42] Z. Nazari, P. Musilek, Impact of digital transformation on the energy sector: a review, Algorithms. 16 (4) (2023) 211. https://www.mdpi.com/1999-4893/16/4/211.
[43] D. Trabucchi, A.S. Patrucco, T. Buganza, G. Marzi, Is transparency the new green? How business model transparency influences digital service adoption, Technovation 126 (2023) 102803, https://doi.org/10.1016/j.technovation.2023.102803.
[44] J.V.V. Lima, W.B. Santos, C. Rodrigues, F. Alencar, Digital Transformation in the public sector: preliminary results of a tertiary literature review, in: 2023 18th Iberian Conference on Information Systems and Technologies (CISTI), 2023.
[45] B. Ly, R. Ly, Emerging trends in social media for E-governance and citizen engagement: a case study of telegram in Cambodia, Comput. Hum. Behav. Rep. 12 (2023) 100347, https://doi.org/10.1016/j.chbr.2023.100347.
[46] I. Mergel, N. Edelmann, N. Haug, Defining digital transformation: results from expert interviews, Gov. Inf. Q. 36 (4) (2019) 101385, https://doi.org/10.1016/j.giq.2019.06.002.
[47] B.W. Wirtz, I. Balzer, D. Schmitt, Mobile Government: research development and research perspectives, Int. J. Public Admin. 46 (4) (2023) 269–290, https://doi.org/10.1080/01900692.2021.1993910.
[48] B.W. Wirtz, P. Daiser, A meta-analysis of empirical e-government research and its future research implications, Int. Rev. Admin. Sci. 84 (1) (2018) 144–163, https://doi.org/10.1177/0020852315599047.
[49] Peña-López, I. (2020). The OECD digital government policy framework. Six dimensions of a digital government.
[50] J.D. Sachs, The Age of Sustainable Development, Columbia University Press, 2015.
[51] M.M. Islam, M. Tareque, Public sector innovation outcome-driven sustainable development in Bangladesh: applying the dynamic autoregressive distributed lag simulations and kernel-based regularised least square machine learning algorithm approaches, J. Public Policy. 43 (2) (2023) 326–357, https://doi.org/10.1017/S0143814x22000368.
[52] K. Spraul, J. Thaler, Partnering for good? An analysis of how to achieve sustainability-related outcomes in public–private partnerships, Bus. Res. 13 (2) (2020) 485–511, https://doi.org/10.1007/s40685-019-0097-3.
[53] S. Baker, Sustainable Development, Routledge, 2016.
[54] A. Vasudevan, B. Wu, S.T. Hai, T. Adialita, S. Johns, A.R.A. Arokiasamy, S. Galdolage, Evaluating the impact of different sustainable leadership: its influence on state-owned enterprises' performance in Northeast China, J. Infrastruct., Policy Develop. 8 (7) (2024) 3909, https://doi.org/10.24294/jipd.v8i7.3909.
[55] A. Yahya, Adnan, T.A. Rozik, S.A Nabil, Green and digital leadership: impact on sustainable performance, mediating environmental knowledge sharing, moderating technological innovation, J. Ecohuman. 3 (7) (2024) 3761–3778, https://doi.org/10.62754/joe.v3i7.4499.
[56] R. Steurer, The role of governments in corporate social responsibility: characterising public policies on CSR in Europe, Policy Sci. 43 (1) (2010) 49–72, https://doi.org/10.1007/s11077-009-9084-4.
[57] H. Kharas, J. McArthur, Building the SDG economy. *Needs, spending, and financing for universal achievement of the sustainable development goals*, Global Econ. Develop. Brookings Working paper 131 (2019) 21.
[58] Burlacu, S., Călin, A.M., Vâlcu, E.C., & Bălu, E.P. (2023). The role of public administration in promoting sustainable development: a case study of population growth and global nutrition interventions. https://doi.org/10.24818/basiq/2023/09/023.
[59] Stiglitz, J.E. (2009). Report by the commission on the measurement of economic performance and social progress.
[60] A. Caro, J. Toro, Effectiveness index for environmental impact assessment methodologies, WIT Trans. Ecol. Environ. 203 (2016) 73–86.
[61] S. Morse, Environmental Performance Index, Routledge, 2019.
[62] K.S. Herman, J. Shenk, Pattern discovery for climate and environmental policy indicators, Environ. Sci. Policy. 120 (2021) 89–98, https://doi.org/10.1016/j.envsci.2021.02.003.
[63] G.K.A. Ribeiro, J. Leitão, Sustainable leadership and sustainable entrepreneurship: a systematic literature review, Discov. Sustain. 5 (1) (2024) 188, https://doi.org/10.1007/s43621-024-00383-3.
[64] B.K. AlNuaimi, S. Kumar Singh, S. Ren, P. Budhwar, D. Vorobyev, Mastering digital transformation: the nexus between leadership, agility, and digital strategy, J. Bus. Res. 145 (2022) 636–648, https://doi.org/10.1016/j.jbusres.2022.03.038.
[65] J. Cui, Does digital strategy, organizational agility, digital lead-ership promote DT? A study of digital strategy, organiza-tional agility, digital leadership affects corporate DT in Chinese technological firms, J. Integr. Soc. Sci. Human. (2024), https://doi.org/10.62836/jissh.v1i1.192.
[66] J.Y.H. Huang, R. Jiang, J.Y.T. Chang, The effects of transformational and adaptive leadership on dynamic capabilities: digital transformation Projects, Project Manag. J. 54 (4) (2023) 428–446, https://doi.org/10.1177/87569728231165896.
[67] L.W. Hooi, A.J. Chan, Does workplace digitalization matter in linking transformational leadership and innovative culture to employee engagement? J. Org. Change Manag. 36 (2) (2023) 197–216, https://doi.org/10.1108/JOCM-06-2022-0184.
[68] C.G. Ugwuja, Transformational leadership in the Digital Transformation era: strategies for effective organizational change, Newport Int. J. Curr. Issues Arts Manag. 4 (2) (2024) 108–113, https://doi.org/10.59298/NIJCIAM/2024/4.2.7108113.
[69] T. Yamin, E. Murwaningsari, Exploring the interplay between digital technology, transformational leadership and agility for enhancing organisational performance, Bus. Ethics Leadership 7 (4) (2023) 73–88, https://doi.org/10.61093/bel.7(4).73-88.2023.
[70] R. Camodeca, A. Almici, Digital transformation and convergence toward the 2030 Agenda's sustainability development goals: evidence from Italian listed firms, Sustainability. 13 (21) (2021) 11831. https://www.mdpi.com/2071-1050/13/21/11831.
[71] C.L. Goi, Digital transformation: a catalyst for achieving sustainable development goals (SDGs) in Malaysia. Perspectives On the Transition Toward Green and Climate Neutral Economies in Asia, IGI Global, 2023, pp. 18–29, https://doi.org/10.4018/978-1-6684-8613-9.ch002.
[72] J. Lin, M. Mao, How does digital transformation affect sustainable innovation performance? The pivotal roles of digital technology-business alignment and environmental uncertainty, Sustain. Develop. 32 (4) (2024) 3163–3181, https://doi.org/10.1002/sd.2830.
[73] W. Yu, Q. Liu, R. Chavez, L. Zheng, Does training provision matter? Unravelling the impact of digital transformation on environmental sustainability, Inf. Technol. People (2023), https://doi.org/10.1108/ITP-01-2023-0007 *ahead-of-print*(ahead-of-print).
[74] A. Kwilinski, O. Lyulyov, T. Pimonenko, Unlocking sustainable value through digital transformation: an examination of ESG performance, Information 14 (8) (2023) 444. https://www.mdpi.com/2078-2489/14/8/444.
[75] Y. Huang, C.-w. Lau, Can digital transformation promote the green innovation quality of enterprises? Empirical evidence from China, PLoS. One 19 (3) (2024) e0296058, https://doi.org/10.1371/journal.pone.0296058.
[76] P.E. Prasetyo, A. Setyadharma, Digitalization technology for sustainable rural entrepreneurship and inequality, J. Human Res. Sustain. Stud. 10 (3) (2022) 464–484, https://doi.org/10.4236/jhrss.2022.103028.
[77] A. Pucihar, M. Kljajić Borštnar, S. Blatnik, R. Bons, in: 37th Bled eConference – Resilience Through Digital Innovation: Enabling the Twin Transition: June 9 –12, 2024, Bled, Slovenia, Conference Proceedings, 2024, https://doi.org/10.18690/um.fov.4.2024.

[78] W. Cao, Z. Cai, X. Yao, L. Chen, Digital transformation to help carbon neutrality and Green sustainable development based on the metaverse, Sustainability. 15 (9) (2023) 7132. https://www.mdpi.com/2071-1050/15/9/7132.
[79] Y. Zhang, S. Jin, How does digital transformation increase corporate sustainability? The moderating role of top management teams, Systems. (Basel) 11 (7) (2023) 355. https://www.mdpi.com/2079-8954/11/7/355.
[80] M. Alzoraiki, A.R. Ahmad, A.A. Ateeq, G.M.A. Naji, Q. Almaamari, B.A.H. Beshr, Impact of teachers' Commitment to the relationship between transformational leadership and sustainable teaching performance, Sustainability. 15 (5) (2023) 4620. https://www.mdpi.com/2071-1050/15/5/4620.
[81] K. Arun, Sustainability in transformational and transactional leadership: turkish textile industry example, in: K. Kankaew, P. Nakpathom, A. Chnitphattana, K. Pitchayadejanant, S. Kunnapapdeelert (Eds.), Applying Business Intelligence and Innovation to Entrepreneurship, IGI Global, 2024, pp. 241–257, https://doi.org/10.4018/979-8-3693-1846-1.ch011.
[82] K.K. Mehmood, M. Rafiq, The impact of transformational leadership climate on business sustainability and new product development process: moderating influence of strategic flexibility, Bull. Bus. Econ. (BBE) 13 (1) (2024), https://doi.org/10.61506/01.00117.
[83] A.O. Ayandibu, Green transformational leadership and organizational behavior effectiveness, in: S.O. Atiku (Ed.), Contemporary Multicultural Orientations and Practices for Global Leadership, IGI Global, 2019, pp. 225–244, https://doi.org/10.4018/978-1-5225-6286-3.ch012.
[84] S.M. Juhro, A.F. Aulia, D. Aliandrina, D. Hadiwaluyo, E. Lavika, The role of catalytic collaboration in leveraging transformational leadership competencies to generate sustainable innovation, 2019.
[85] N.A. Zhuravleva, Transformational leadership and sustainable development of Russian transport systems, Modern Transport. Syst. Technol. 8 (2) (2022) 92–111, https://doi.org/10.17816/transsyst20228292-111.
[86] H.M. Naguib, A.E.-H.M.A. Naem, The impact of transformational leadership on the organizational innovation, Int. J. Soc. Sci. Human. Invent. 5 (1) (2018) 4337–4343, https://doi.org/10.18535/ijsshi/v5i1.15.
[87] S. Suwanto, D. Sunarsi, W. Achmad, Effect of transformational leadership, servant Leadershi, and digital transformation on MSMEs performance and work innovation capabilities, CEMJP 30 (4) (2022) 751–762. http://journals.kozminski.cem-j.org/index.php/pl_cemj/article/view/76.
[88] M. Asif, L. Yang, M. Hashim, The role of digital transformation, corporate culture, and leadership in enhancing corporate sustainable performance in the manufacturing sector of China, Sustainability. 16 (7) (2024) 2651. https://www.mdpi.com/2071-1050/16/7/2651.
[89] J.W. Creswell, J.D. Creswell, Research design: Qualitative, quantitative, and Mixed Methods Approaches, Sage publications, 2017.
[90] J.F. Hair, J.J. Risher, M. Sarstedt, C.M. Ringle, When to use and how to report the results of PLS-SEM, Eur. Bus. Rev. 31 (1) (2019) 2–24, https://doi.org/10.1108/EBR-11-2018-0203.
[91] M. Sarstedt, C.M. Ringle, J.F. Hair, Partial least squares structural equation modeling, in: C. Homburg, M. Klarmann, A. Vomberg (Eds.), Handbook of Market Research, Springer International Publishing, 2022, pp. 587–632, https://doi.org/10.1007/978-3-319-57413-4_15.
[92] I. Etikan, K. Bala, Sampling and sampling methods, Biom. Biostat. Int. J. 5 (6) (2017) 00149.
[93] H. Taherdoost, Sampling methods in research methodology; how to choose a sampling technique for research, Int. J. Acad. Res. Manag. (IJARM) 5 (2016).
[94] D.S. Soper. A-priori sample size calculator for structural equation models [Software], 2024. https://www.danielsoper.com/statcalc.
[95] S.A. Carless, A.J. Wearing, L. Mann, A short measure of transformational leadership, J. Bus. Psychol. 14 (3) (2000) 389–405, https://doi.org/10.1023/A:1022991115523.
[96] M. Olczyk, M. Kuc-Czarnecka, Digital transformation and economic growth-DESI improvement and implementation, Technol. Econ. Develop. Econ. 28 (2022) 775–803, https://doi.org/10.3846/tede.2022.16766.
[97] Lincaru, C. (2021). Measuring and delivering social inclusion: leave no one behind in the digital transformation journey.
[98] A. Ramirez, E. Inga, Educational innovation in adult learning considering digital transformation for Social inclusion, Educ. Sci. (Basel) 12 (12) (2022) 882. https://www.mdpi.com/2227-7102/12/12/882.
[99] A.K. Feroz, H. Zo, A. Chiravuri, Digital transformation and environmental sustainability: a review and research agenda, Sustainability. 13 (3) (2021) 1530. https://www.mdpi.com/2071-1050/13/3/1530.
[100] S.u. Rehman, A.R. Gill, Digital Transformation, Institutional quality, and environmental sustainability: evidence from SAARC countries, iRASD J. Econ. 5 (2) (2023) 392–407, https://doi.org/10.52131/joe.2023.0502.0135.
[101] M. Nasiri, J. Ukko, M. Saunila, T. Rantala, Managing the digital supply chain: the role of smart technologies, Technovation 96-97 (2020) 102121, https://doi.org/10.1016/j.technovation.2020.102121.
[102] R.J. Little, D.B. Rubin, Statistical Analysis with Missing Data, 793, John Wiley & Sons, 2019.
[103] J.F. Hair, G.T.M. Hult, C.M. Ringle, M. Sarstedt, A Primer On Partial Least Squares Structural Equation Modeling (PLS-SEM), SAGE Publications, 2022. Incorporated.
[104] H.F. Kaiser, J. Rice, Little Jiffy, Mark Iv, Educ. Psychol. Meas. 34 (1) (1974) 111–117, https://doi.org/10.1177/001316447403400115.
[105] J.F. Hair, C.M. Ringle, M. Sarstedt, PLS-SEM: indeed a silver bullet, J. Mark. Theory Practice 19 (2) (2011) 139–152, https://doi.org/10.2753/MTP1069-6679190202.
[106] P.M. Podsakoff, S.B. MacKenzie, N.P. Podsakoff, Sources of method bias in social science research and recommendations on how to control it, Annu. Rev. Psychol. 63 (2012) 539–569, https://doi.org/10.1146/annurev-psych-120710-100452. Volume 63, 2012.
[107] L.M. Molina, J. Lloréns-Montes, A. Ruiz-Moreno, Relationship between quality management practices and knowledge transfer, J. Operat. Manag. 25 (3) (2007) 682–701, https://doi.org/10.1016/j.jom.2006.04.007.
[108] Z. Awang, Structural Equation Modeling Using AMOS Graphic, Penerbit Universiti Teknologi MARA, 2012.
[109] B.J. Zaini, Z. Md Yusof, R. Mansor, N. Zainuddin, N.F. Khairul Adlee, H.N. Lop Ahmad, Constructing motivation for sports activities among college students using CB-SEM path model, Eur. J. Mol. Clin. Med. 7 (6) (2020) 2299–2313.
[110] C. Fornell, D.F. Larcker, Structural equation models with unobservable variables and measurement error: algebra and statistics, J. Mark. Res. 18 (3) (1981) 382–388.
[111] J. Henseler, C.M. Ringle, M. Sarstedt, A new criterion for assessing discriminant validity in variance-based structural equation modeling, J. Acad. Mark. Sci. 43 (1) (2015) 115–135, https://doi.org/10.1007/s11747-014-0403-8.
[112] D. Gallardo-Vázquez, M. Isabel Sánchez-Hernández, Structural analysis of the strategic orientation to environmental protection in SMEs, BRQ Bus. Res. Quarterly 17 (2) (2014) 115–128, https://doi.org/10.1016/j.brq.2013.12.001.
[113] J. Henseler, G. Hubona, P.A. Ray, Using PLS path modeling in new technology research: updated guidelines, Ind. Manag. Data Syst. 116 (1) (2016) 2–20.
[114] L.t. Hu, P.M. Bentler, Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives, Struct. Eq. Model.: Multidiscipl. J. 6 (1) (1999) 1–55, https://doi.org/10.1080/10705519909540118.
[115] J.F. Hair, G.T.M. Hult, C.M. Ringle, M. Sarstedt, N.P. Danks, S. Ray, Partial Least Squares Structural Equation Modeling (PLS-SEM) Using R: A Workbook, Springer Nature, 2021.
[116] E. Kara, G. Kirpik, A. Kaya, A research on digital violence in social Media, in: F. Özsungur (Ed.), Handbook of Research on Digital Violence and Discrimination Studies, IGI Global, 2022, pp. 270–290, https://doi.org/10.4018/978-1-7998-9187-1.ch013.
[117] R.L. Worthington, T.A. Whittaker, Scale development research:a content analysis and recommendations for best practices, Couns. Psychol. 34 (6) (2006) 806–838, https://doi.org/10.1177/0011000006288127.
[118] R.F. Falk, N.B. Miller, A Primer for Soft Modeling, University of Akron Press, 1992.
[119] I. Castro, J.L. Roldán, A mediation model between dimensions of social capital, Int. Bus. Rev. 22 (6) (2013) 1034–1050, https://doi.org/10.1016/j.ibusrev.2013.02.004.
[120] J. Cohen, Statistical Power Analysis for the Behavioral Sciences, 2nd ed., Routledge, 1998 https://doi.org/10.4324/9780203771587.
[121] X. Zhao, J.G. Lynch Jr., Q. Chen, Reconsidering Baron and Kenny: myths and truths about mediation analysis, J. Consum. Res. 37 (2) (2010) 197–206, https://doi.org/10.1086/651257.